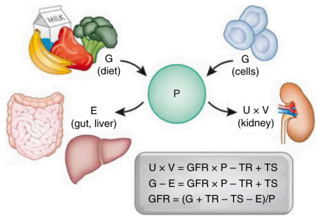

Fig. 23.1 Determinants of the serum level of endogenous filtration markers. The plasma level *(P)* of an endogenous filtration marker is determined by its generation *(G)* from cells and diet, extrarenal elimination *(E)* by gut and liver, and urinary excretion *(UV)* by the kidney. Urinary excretion is the sum of filtered load (glomerular filtration rate *[GFR]* × *P*) minus reabsorption *(TR)* plus tubular secretion *(TS)*. In the steady state, generation minus extrarenal elimination equals urinary excretion. By substitution and rearrangement, GFR can be expressed as the ratio of the non-GFR determinants (*G, TS, TR,* and *E*) to the plasma level. (Permission granted from *J Am Soc Nephrol* for Stevens LA, Levey AS. Measured GFR as a confirmatory test for estimated GFR. *J Am Soc Nephrol.* 2009;20(11):2305–2313.)

Fig_23_1

Day	P_{marker}	eGFR
0	1.0	135
1.5	1.6	79
2.0	1.8	69
2.5	1.9	65
3.0	2.0	60

Fig. 23.2 Effect of an acute GFR decline on generation, filtration, excretion, balance, plasma concentration of an endogenous filtration marker (P_{marker}), and estimated GFR *(eGFR)* based on the marker. After an acute GFR decline, generation of the marker is unchanged, but filtration and excretion are reduced, resulting in retention of the marker (a rising positive balance), a rising plasma concentration, and declining eGFR (non–steady state). During this time, eGFR is higher than GFR. While GFR remains reduced, the rise in plasma concentration leads to an increase in filtered load (the product of GFR times the plasma level) until filtration equals generation. At that time, cumulative balance and the plasma level plateau at a new steady state. In the new steady state, eGFR approximates mGFR. GFR is expressed in units of milliliter per minute per 1.73 m^2. Tubular secretion and reabsorption and extrarenal elimination are assumed to be zero. (Permission granted from *J Am Soc Nephrol* for Stevens LA, Levey AS. Measured GFR as a confirmatory test for estimated GFR. *J Am Soc Nephrol.* 2009;20(11):2305–2313.)

Fig_23_2

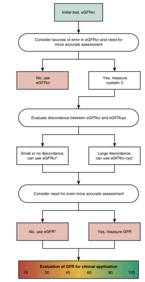

Fig. 23.3 Glomerular filtration rate (GFR) evaluation using initial and supportive tests. The algorithm describes the approach to the evaluation of GFR. Our approach is to use initial and supportive testing to develop a final assessment of true GFR and to apply it in individual decision making at each point in time. The initial test for evaluation of GFR is often *eGFRcr*, which will be available in most patients because creatinine is measured routinely as part of the basic metabolic panel. If the eGFRcr is expected to be inaccurate, or if a more accurate assessment of GFR is needed for clinical decision making, such as diagnosis or staging of CKD or drug dosing, then cystatin C should be measured and the discordance between eGFRcr and *eGFRcys* should be assessed. If eGFRcr and eGFRcys are not discordant (within 15 mL/min/1.73 m^2 or <20%–30% of each other), then accuracy of eGFRcr, eGFRcys, and eGFRcr-cys is similar. If eGFRcr and eGFRcys are discordant (not within 15 mL/min/1.73 m^2 or > 20%–30% of each other), then eGFRcr-cys is generally more accurate than either eGFRcr or eGFRcys, with some exceptions, such as otherwise healthy populations with increased creatinine generation owing to increased muscle mass, or decreased creatinine secretion

Fig_23_3

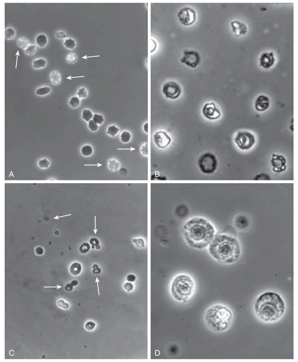

Fig. 23.4 Erythrocytes in urine. (A) Isomorphic erythrocytes, some with a "crenated" appearance *(arrows)*. (B) Different types of dysmorphic erythrocytes. (C) Acanthocytes *(arrows)*. (D) Proximal renal tubule cells. (From Fogazzi GB, Verdesca S, Garigali G. Urinalysis: core curriculum 2008. *Am J Kidney Dis.* 2008;51:1052–1067.)

Fig_23_4

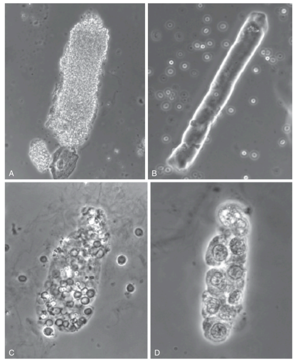

Fig. 23.5 Urinary casts. (A) Finely granular cast. (B) Waxy cast with the typical "melted wax" appearance. (C) Red blood cell cast. (D) Renal tubule epithelial cell cast. (From Stevens LA, Nolin TD, Richardson MM, et al. Comparison of drug dosing recommendations based on measured GFR and kidney function estimating equations. *Am J Kidney Dis.* 2009;54:33–42.)

Fig_23_5

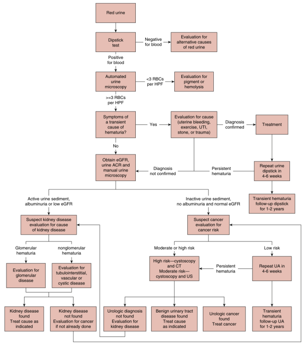

Fig. 23.6 Evaluation of hematuria. The algorithm describes our approach to the evaluation of hematuria. Hematuria should first be confirmed via dipstick and automated urine microscopy, followed by evaluation for transient causes of hematuria (such as uterine bleeding, exercise, urinary tract infection, stone, or trauma). The patient should then receive a stepwise evaluation for kidney disease, followed by risk stratification/evaluation for bladder cancer and urologic causes of hematuria. High-risk patients are ≥60 years old and have >30–pack-year smoking history, >25 red blood cells/HPF on a single urine microscopy, or history of gross hematuria. Moderate-risk patients are women 50–59 years old, men 40–59 years old, people with a 10– to 30–pack-year history of smoking, those with 11 to 25 RBC/HPF on a single urine microscopy, and other relevant history. Negative workups for cancer or urologic disease should prompt a formal workup for kidney disease and vice versa.

Fig_23_6

Table 23.1 Clinical applications of laboratory assessment of kidney disease

Topic/Application	Population	GFR	Proteinuria	Dipstick	Urine microscopy
Significance	All	Overall index of kidney function	Albuminuria as a marker of kidney damage	Initial test for markers of kidney damage	Confirmatory test for markers of kidney damage
Defining kidney disease	AKI	↑Scr by 0.3 mg/dL in 48 hours or by 50% in 7 days, or oliguria >4 hours	NA	NA	NA
	AKD	AKI or GFR <60 or ↓GFR by 35% for <3 months	ACR >30 for <3 months	Present for <3 months	Present for <3 months
	CKD	GFR <60 for >3 months	ACR >30 for >3 months	Present for >3 months	Present for >3 months
Staging kidney disease (severity)	AKI	Stages 1-3 defined by changes in Scr, urine output, and initiation of KRT	NA	NA	NA
	AKD	As for AKI if AKI present As for CKD if AKI absent	NA if AKI present As for CKD if AKI absent	NA	NA
	CKD	G1-G5 defined by level of GFR	A1-A3 defined by level of albuminuria	NA	NA
Screening special populations/detecting kidney disease in people at increased risk	AKI AKD CKD	Yes Yes Yes	NA Yes Yes	NA Yes Yes	NA Yes Yes
Diagnosing the cause of kidney disease	AKI/AKD/CKD	NA	Yes	Yes	Yes
Adjusting drug dosing in kidney disease	AKI/AKD/CKD	Yes	NA	NA	NA
Assessing progression of kidney disease	AKI AKD/CKD	Yes Yes	NA Yes	NA NA	NA Yes

AKD, acute kidney disease; AKI, acute kidney injury; CKD, chronic kidney disease; GFR, glomerular filtration rate in mL/min/1.73 m^2; GP, general population; KRT, kidney replacement therapy; NA, not applicable; Yes, applicable.
Markers of kidney damage include albuminuria, urinalysis abnormalities, imaging abnormalities, pathologic abnormalities, and electrolyte disorders due to altered tubular function. Only albuminuria, urine dipstick, and urine microscopy are included in this chapter.

Table_23_1

Table 23.2 Examples of conditions in which eGFR using creatinine or cystatin C are less accurate

Use	eGFRcr[a]	eGFRcys[a]
Nonsteady state (AKI)	Change in eGFR lags behind the change in mGFR (eGFR overestimates mGFR when mGFR is declining and underestimates mGFR when mGFR is rising)	Change in eGFR lags behind the change in mGFR (eGFR overestimates mGFR when mGFR is declining and underestimates mGFR when mGFR is rising)
Factors affecting generation[b]	Decreased by large muscle mass, high protein diet, ingestion of cooked meat, and creatine supplements Increased by small muscle mass, limb amputation, muscle-wasting diseases	Presumed decreased in greater adiposity, smoking, hyperthyroidism, glucocorticoid excess, and chronic inflammation, as indicated by insulin resistance, high levels of C-reactive protein and tumor necrosis factor, or low levels of serum albumin Presumed increased in hypothyroidism
Factors affecting tubular reabsorption or secretion[b]	Decreased by drug-induced inhibition of secretion	NA
Factors affecting extrarenal elimination[b]	Decreased by inhibition of gut creatininase by antibiotics Increased by dialysis, large losses of extracellular fluid (drainage of pleural fluid or ascites)	Increased by large losses of extracellular fluid (drainage of pleural fluid or ascites)
Range	Less precise at higher GFR, due to higher biological variability in non-GFR determinants relative to GFR and larger measurement error in Scr and GFR	Less precise at higher GFR, due to higher biological variability in non-GFR determinants relative to GFR, and larger measurement error in Scys and GFR
Interference with assays	Spectral interferences (bilirubin, some drugs) Chemical interferences (glucose, ketones, bilirubin, some drugs)	At high levels, may be susceptible to hook effects

[a]Effects eGFR, opposite effects on serum creatinine and serum cystatin C.
[b]Generation, tubular reabsorption and secretion, and extrarenal elimination are non-GFR determinants. Errors in eGFRcr-cys related to non-GFR determinants of creatinine or cystatin are hypothesized to be less than for eGFRcr and eGFRcys alone.
AKI, Acute kidney injury; *eGFRcr,* glomerular filtration rate estimates based on serum creatinine; *eGFRcys,* glomerular filtration rate estimates based on cystatin C; *GFR*, glomerular filtration rate; *mGFR,* measured glomerular filtration rate; *Scr,* serum creatinine; *Scys,* serum cystatin C.

Table_23_2

Table 23.3 Most commonly used validated equations for estimating glomerular filtration rate

Marker	Equation name and year	Age	Variables	Development populations
Creatinine	CKD-EPI 2009[191,192]	≥18; Modification CKD-EPI 40 for pediatric available	Developed using ASR but reported not using Black race coefficient, ASR (NB)	8254 Black and non-Black individuals from 10 studies in the United States and Europe[a]
	CKD-EPI 2021[68]	≥18	AS	8254 Black and non-Black individuals from 10 studies in the United States and Europe[a]
	EKFC 2021[6]	2-100	AS, European Black and non-Black specific Q-value; Separate Q-values for Africa vs. Europe	mGFR vs. SCr (11,251 participants in 7 studies in Europe and 1 study from United States) Normal GFR from 5482 participants in 12 studies of kidney donor candidates 100% Caucasian) European non-Black Q from 83,157 laboratory samples (age 2-40 years) in 3 European hospital clinical laboratories; European Black Q-value (*N* = 90 living kidney donors from Paris); African Black Q-value (*N* = 470 healthy individuals from République Démocratique de Congo); All Q-values developed in cohorts independent for EKFC development and validation
	Lund-Malmo Revised 2011[193]		AS	3495 GFR examinations from 2847 adults from Sweden referred for measurement of GFR
	CKD-EPI 2009 Modified for China 2014[b]	≥18	AS	589 People with diabetes from X Third Affiliated Hospital of Sun Yat-sen
	CKD-EPI 2009 Modified for Japan 2016[b]	≥18	AS	413 Hospitalized Japanese people in 80 medical centers
	CKD-EPI 2009 Modified for Pakistan 2013[b,194]	≥18	AS	542 Randomly selected low- to middle-income communities in Karachi and 39 people from the kidney clinic
	CKiD U25 2021[83]	1–25	AS, height	928 children with CKD in the United States
Cystatin C	CKD-EPI 2012[195]	≥18	AS	5352 Black and non-Black individuals from 13 studies in the United States and Europe
	CKD-EPI 2023[63]	≥18	A	5352 Black and non-Black individuals from 13 studies in the United States and Europe
	EKFC 2023[5]	2-100	A	mGFR vs. SCys (assumed to be the same as mGFR vs. SCr) Normal GFR (same as for SCr equation) Q from laboratory samples from 227,643 (42% female) laboratory samples from Uppsala University Hospital, Sweden
	CAPA 2014[196]	2-86	AS	4690 Individuals within large subpopulations of children and Asian and Caucasian adults
	CKiD U25 2021[83]	1-25	AS	928 Children with CKD in United States
Creatinine-cystatin C	CKD-EPI 2012[195]	≥18	Developed using ASR but reported not using Black race coefficient, ASR (NB)	5352 Black and non-Black individuals from 13 studies in the United States and Europe
	CKD-EPI 2021[68]	≥18	AS	5352 Black and non-Black individuals from 13 studies in the United States and Europe

[a]Also included 100 Asians and 353 Hispanic or Native Americans.
[b]Modified from CKD-EPI or MDRD; modifications may reflect systematic differences in measurement of creatinine and mGFR, as well as population differences in nonGFR determinants of creatinine. To obtain eGFRcr-cys, one can average LMR 2011 and CAPA 2014; CKid U25 creatinine and cystatin C; or EKFC creatinine 2021 and cystatin C 2023 equations.
A, Age; *CAPA*, Caucasian and Asian pediatric and adult subjects; *CKD-EPI*, Chronic Kidney Disease Epidemiology Collaboration; *CKiD*, Chronic kidney disease in children; *cr*, creatinine; *cys*, cystatin C; *EKFC*, European Kidney Function Consortium; *GFR*, glomerular filtration rate; *mGFR*, measured glomerular filtration rate; *NB*, non-Black; *R*, race; *S*, sex; *SCr*, serum creatinine; *SCys*, serum cystatin C.
From KDIGO. 2024 Clinical Practice Guideline for the Evaluation and Management of Chronic Kidney Disease. Stevens, Paul E. et al. *Kidney Int.* 2024;105(4):S117–S314.

Table_23_3

Table 23.4 Kidney disease: Improving global outcomes (KDIGO) guideline: categories of proteinuria

	Normal to mildly increased (A1)	Moderately increased (A2)	Severely increased (A3)
Timed urine collection			
AER (mg/24 h)	<30	30-300	>300
PER (mg/24 h)	<150	150-500	>500
Spot or timed collection			
ACR:			
mg/mmol	<3	3-30	>30
mg/g	<30	30-300	>300
PCR:			
mg/mmol	<15	15-50	>50
mg/g	<150	150-500	>500
Protein reagent strip	Negative to trace	Trace to +	+ or greater

Relationships between AER and ACR and between PER and PCR are based on the assumption that average creatinine excretion rate is 1.0 g/day or 10 mmol/day (conversions are rounded for pragmatic reasons). ACR and PCR may also be reported as μg/mg (1 μg/mg = 1 mg/g) or mg/mmol (1 mg/mmol = 0.001 mg/g)
ACR, Albumin-to-creatinine ratio; *AER,* albumin excretion rate; *PCR,* protein-to-creatinine ratio; *PER,* protein excretion rate.
From Kidney Disease: Improving Global Outcomes (KDIGO) CKD Work Group. KDIGO 2012 clinical practice guideline for the evaluation and management of chronic kidney disease. *Kidney Int Suppl*. 2013;3:1–150.

Table_23_4

Table 23.5 Evaluation of proteinuria in adults

Initial testing of proteinuria (in descending order of preference, in all cases an early morning urine sample is preferred)
- Urine albumin-to-creatinine ratio (ACR)
- Urine protein-to-creatinine ratio (PCR)
- Reagent strip urinalysis for total protein with automated reading
- Reagent strip urinalysis for total protein with manual reading

Confirmatory testing for proteinuria
- Confirm reagent strip positive albuminuria and proteinuria by quantitative laboratory measurement and express as a ratio to creatinine wherever possible.
- Confirm ACR ≥30 mg/g on a random untimed urine with a subsequent early morning urine sample.
- If a more accurate estimate of albuminuria or total proteinuria is required, measure AER or PER in a timed urine sample

Clinical laboratory reporting of ACR and PCR in untimed urine samples in addition to albumin concentration or proteinuria concentrations rather than concentrations alone.
From Kidney Disease: Improving Global Outcomes (KDIGO) CKD Work Group. KDIGO 2012 clinical practice guideline for the evaluation and management of chronic kidney disease. *Kidney Int Suppl*. 2013;3:1–150.

Table_23_5

Table 23.6 Biological, preanalytical, and analytical factors that may impact variability in urinary protein or albumin or ACR

Biological	
Increase in albumin or protein	Hematuria High-intensity exercise Uterine bleeding Posture (postural proteinuria)
Increase in protein	Urinary tract infection or inflammation
Increase in ACR or PCR due to lower urine creatinine	Older age Female sex Reduced muscle mass Decreased dietary protein intake Decreased physical activity
Preanalytical	
Collection type	Timed or untimed (spot)
Time of day	First morning vs. random void (first morning reduces variation)
Storage conditions	Degradation of protein or albumin during storage Adsorption to plastic Storage temperature
Analytical	
	Total protein or albumin measurement High albumin levels can result in high-dose hook effect

Adapted from Johnson DW, Jones GR, Mathew TH, et al. Chronic kidney disease and measurement of albuminuria or proteinuria: a position statement. *Med J Aust*. 2012;197:224–225; and Miller WG, Bruns DE, Hortin GL, et al. Current issues in measurement and reporting of urinary albumin excretion [article in French]. *Ann Biol Clin (Paris)*. 2010;68:9–25.

Table_23_6

Table 23.7 Main causes of abnormal color changes in urine

	Cause	Color
Pathologic conditions	Macroscopic hematuria, hemoglobinuria, myoglobinuria	Pink, red, brown, black
	Jaundice	Yellow to brown
	Chyluria	White milky
	Massive uric acid crystalluria	Pink
	Porphyrinuria, alkaptonuria	Red to black; increases after urine left to stand
Medications	Rifampin	Yellow-orange to red
	Propofol	White
	Phenytoin, phenazopyridine	Red
	Chloroquine, nitrofurantoin	Brown
	Triamterene, blue dyes of enteral feeds	Green
	Metronidazole, methyldopa, imipenem–cilastatin	Darkening after urine left to stand
Foods	Beetroot	Red
	Senna rhubarb	Yellow to brown-red

From Fogazzi GB, Verdesca S, Garigali G. Urinalysis: core curriculum 2008. *Am J Kidney Dis*. 2008;51:1052–1067; and Davsion A. *Urinalysis*. 3rd ed. Oxford: Oxford University Press; 2005.

Table_23_7

Table 23.8 Causes of hematuria

		Urine microscopy findings				
Location in the urinary tract	Clinical entity	RBC casts or dysmorphic RBCs	WBC	Crystals	Albuminuria	Symptoms
Kidney						
Glomeruli	Primary glomerular disease (e.g., Alport syndrome, IgA, C3, and dense deposit disease)	+	–	–	±	–
	Secondary glomerular disease (e.g., lupus glomerulonephritis, vasculitis, and postinfectious glomerulonephritis)	+	+	–	+	+
Tubules, interstitium, vessels, cysts	Papillary necrosis	–	±	–	±	+
	Sickle cell disease	–	±	–	±	±
	Autosomal dominant polycystic kidney disease	–	±	–	±	±
	Infarct (thrombosis, embolisms, dissection)	–	±	–	±	±
Upper urinary tract	Benign kidney masses	–	±	–	–	–
	Malignancy	–	±	–	–	±
	Pyelonephritis	–	+	–	–	+
	Nephrolithiasis					
Lower urinary tract	Malignancy (bladder, ureter, prostate)	–	±	–	–	±
	Ureterolithiasis	–	+	±	–	±
	BPH	–	–	–	–	±
	Urinary tract infection	–	+	–	–	+
Any region of the urinary tract	Exercise hematuria	±	+	–	±	±
	Bleeding diathesis or anticoagulation	–	–	–	–	±
	Hypercalciuria and hyperuricosuria	–	±	±	–	–
	Drugs (cyclophosphamide or crystal-inducing drugs)	–	–	±	–	±
	Trauma	–	–	–	–	+

D, Dysmorphic; *I,* isomorphic; + almost always present; ± sometimes present; – rarely present.

Table_23_8

Table 23.9 Common casts

Cast	Main clinical association(s)
Hyaline	Normal and in kidney disease
Fine granular	Normal and in kidney disease; casts contain protein aggregates
Coarse granular	Kidney disease of any cause; casts contain degenerated cellular elements
Waxy	Kidney disease of any cause; casts result from degeneration of other casts
Broad	Chronic kidney disease; casts reflect hypertrophy of the tubule
Fatty	Kidney disease with heavy proteinuria
Red blood cell	Proliferative glomerulonephritis; casts result from glomerular bleeding
White blood cell	Inflammatory tubulointerstitial disease
Tubular epithelial cell	Often the result of active acute kidney diseases such as acute tubular necrosis, proliferative glomerulonephritis, acute interstitial nephritis, thrombotic microangiopathy; casts formed from the aggregation of desquamated cells of the tubule

From Fogazzi GB, Verdesca S, Garigali G. Urinalysis: core curriculum 2008. *Am J Kidney Dis*. 2008;51:1052–1067.

Table_23_9

Table 23.10 Common crystals and their appearance

Crystal	Appearance	pH range	Birefringence
Endogenous			
Ammonium biurate	Yellow-brown spheres with spicules, thorn apples	5.5-7.0	Strong
Bilirubin crystals	Yellow needle-like crystals, attach to cell surfaces	5.5	Weak
Calcium carbonate	Dumbbells, thick rods, 4-leaf clover	7.0	Strong
Calcium oxalate	Monohydrate: colorless ovoid, dumbbells, rods	5.4-6.7	Strong
	Dihydrate: colorless bipyramidal	5.4-6.7	Weak
Calcium phosphate	Prisms, sticks, needles, stars, rosettes in isolation or in aggregates	6.7-7.0	Strong
Cholesterol	Thin plates with well-defined edges	5.5	Negative
Cystine	Colorless hexagonal plates with irregular sides	5.5	Weak
2,8-Dihydroxyadenine	Reddish-brown round with central spicules and dark outline	5.5-7.0	Maltese cross
Leucine	Yellow-brown spheres with concentric striations	5.5-6.5	Maltese cross
Tyrosine	Colorless to yellow thin needles in bundles or rosettes	5.5-6.5	Strong
Triple phosphate (struvite)	Trapezoids, prisms, feather-like, "coffin lids"	6.2-7.0	Strong
Drug-related			
Acyclovir	Thin needles with sharp or blunt ends	5.5-7.0	Strong
Amoxicillin	Colorless thin needles, broom/brushlike	5.5-6.5	Strong
Atazanavir	Thin needles in isolation or as aggregates	6.0-7.0	Strong
Ciprofloxacin	Colorless needles, stars, fans, sheaves	>7.0	Strong
Methotrexate	Yellow-brown	5.4-6.0	Strong
Sulfadiazine	Amber as shocks or sheaves of wheat, shells	5.5	Strong
Triamterene	Brown and other colors (green/orange/red); spheres	5.5	Maltese cross
Vitamin C (calcium oxalate)	Same as for monohydrated calcium oxalate	5.4-6.7	Strong

From Cavanaugh C, Perazella MA. Urine sediment examination in the diagnosis and management of kidney disease: core curriculum 2019. *Am J Kidney Dis*. 2019;73(2):258–272.

"macroalbuminuria" or "clinical proteinuria," referring to dipstick positive proteinuria, approximately equivalent to A3, are no longer recommended.) Nephrotic-range proteinuria refers to protein excretion of >3.5 g/24 hours. Nephrotic syndrome is defined as urine total protein excretion >3000 or 3500 mg/day, hypoalbuminemia, hyperlipidemia, and edema.

eGFRdiff (eGFRcr lower than eGFRcys) is associated with a lower risk of adverse outcomes. These findings suggest that the difference in risk of adverse outcomes is likely due to differential confounding of the relationship of GFR to adverse outcomes by factors associated with the non-GFR determinants of the markers (positive confounding by high eGFRcys vs. negative confounding by high eGFRcr).

Table_23_10

Table 23.11 Use of laboratory assessment in diagnosis of cause of kidney disease

Kidney parenchymal involvement	Presence of "active disease" on pathologic examination(examples below)	GFR	Proteinuria	Dipstick	Possible microscopy findings
None	No	NA	A1	—	—
Glomerular diseases	Yes (proliferation)	NA	A3 (may be nephrotic range)	Heme	Dysmorphic RBC, RBC casts, RTE, granular casts
	No	NA	A3 (may be nephrotic range)		Lipid, RTE, granular casts
Tubulointerstitial disease	Yes (inflammation)	NA	A2/A3 (may have nonalbumin proteinuria)	Leukocyte esterase	WBC and WBC casts, nondysmorphic RBC, RTE, granular casts
	No	NA	A1/A2 (may have nonalbumin proteinuria)	May have defects in urine concentration and acidification	RTE, granular casts
Vascular diseases	Yes (thrombotic microangiopathy)	NA	A2/A3	Heme	Nondysmorphic RBC, RTE, granular casts
	No	NA	A1/A2		RTE, granular casts
Other	No	NA	A1/A2		RTE, granular casts

None includes decreased kidney perfusion or acute obstruction. Nonproliferative glomerular diseases include diabetic nephropathy, podocytopathies, and membranous nephropathy. Inflammatory tubulointerstitial diseases include kidney diseases associated with urinary tract disorders including infection, stones, and chronic obstruction. Other refers to diseases that may affect any location in the kidney parenchyma, such as cystic kidney diseases or congenital anomalies of the kidneys and urinary tracts (CAKUT). Genetic diseases are not considered a separate category because diseases in each category are now recognized as having genetic determinants. Drug toxicity is not considered a separate category but can affect all locations in the kidney parenchyma. Pathologic findings in kidney transplant recipients are not fundamentally different from those in native kidney disease, except for the occurrence of acute and chronic tubulointerstitial and vascular rejection.
NA, Not applicable as level of GFR is generally not helpful in ascertaining cause of disease. Exceptions include minimal change disease, in which GFR is normal, and acute tubular necrosis and tubulointerstitial transplant rejection, in which GFR is reduced. Proteinuria, dipstick, and microscopy categories reflect most likely findings; individual patients may not fit into these categories. A1, A2, and A3 refer to categories for albuminuria.
RBC, Red blood cell; *RTE,* renal tubular epithelial cell; *WBC,* white blood cell.

Table_23_11